Supplementary Tables

Anonymous Author(s)

This document contains the tables cited as Supplementary Tables S1–S12 in the anonymous manuscript. The numbering is independent of the tables retained in the 12-page paper.

Table S1: Dataset statistics under the absolute-time sliding-window temporal cold-start protocol. Each dataset is divided into five chronological windows $W_1, \ldots, W_5$. We evaluate on Test Window W_3, W_4, and W_5, where each test window uses its preceding windows for training and validation. Cold ratio denotes the cold-item proportion in the corresponding test split.

Dataset	Split	Train window	Test window	Train sequences	Val sequences	Test sequences	Test targets	Cold ratio
Test Window W_3								
Beauty	1	W_1	W_3	12	5	56	254	0.984
Sports	1	W_1	W_3	3	2	37	131	1.000
Toys	1	W_1	W_3	101	57	324	900	0.903
Sephora	1	W_1	W_3	5,091	725	1,454	2,917	0.634
WikiLife	1	W_1	W_3	509	36	113	286	0.650
WeiboDaily	1	W_1	W_3	854	506	909	37,147	0.285
WeiboTech	1	W_1	W_3	134	63	101	14,764	0.469
Test Window W_4								
Beauty	2	$W_{1:2}$	W_4	86	56	794	2,828	0.957
Sports	2	$W_{1:2}$	W_4	69	37	641	2,379	0.960
Toys	2	$W_{1:2}$	W_4	684	324	859	3,642	0.901
Sephora	2	$W_{1:2}$	W_4	12,523	1,454	5,337	14,274	0.823
WikiLife	2	$W_{1:2}$	W_4	1,650	113	451	1,348	0.593
WeiboDaily	2	$W_{1:2}$	W_4	1,597	909	1,233	59,954	0.247
WeiboTech	2	$W_{1:2}$	W_4	216	101	212	21,740	0.409
Test Window W_5								
Beauty	3	$W_{1:3}$	W_5	1,043	794	4,434	36,100	0.904
Sports	3	$W_{1:3}$	W_5	904	641	6,836	40,569	0.883
Toys	3	$W_{1:3}$	W_5	1,538	859	3,959	31,265	0.925
Sephora	3	$W_{1:3}$	W_5	27,351	5,337	21,944	84,616	0.863
WikiLife	3	$W_{1:3}$	W_5	5,152	451	4,339	10,088	0.463
WeiboDaily	3	$W_{1:3}$	W_5	2,457	1,233	1,775	99,701	0.228
WeiboTech	3	$W_{1:3}$	W_5	438	212	405	41,099	0.277

Table S2: Detailed @5 performance of TIGER under temporal cold-start evaluation.

Dataset	Cold ratio	Seen			Unseen		
		H@5	R@5	N@5	H@5	R@5	N@5
Test Window W_3							
Beauty	0.984	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Sports	1.000	–	–	–	0.00000	0.00000	0.00000
Toys	0.903	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Sephora	0.634	0.16915	0.19306	0.13842	0.00000	0.00000	0.00000
WikiLife	0.650	0.38298	0.35000	0.20709	0.00000	0.00000	0.00000
WeiboDaily	0.285	0.59873	0.22879	0.17338	0.00000	0.00000	0.00000
WeiboTech	0.469	0.23171	0.00243	0.00097	0.00000	0.00000	0.00000
Test Window W_4							
Beauty	0.957	0.02439	0.00000	0.00000	0.00000	0.00000	0.00000
Sports	0.960	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Toys	0.901	0.02703	0.00557	0.00240	0.00000	0.00000	0.00000
Sephora	0.823	0.09255	0.14625	0.11730	0.00000	0.00000	0.00000
WikiLife	0.593	0.28244	0.18613	0.13446	0.00000	0.00000	0.00000
WeiboDaily	0.247	0.70188	0.28315	0.21476	0.00000	0.00000	0.00000
WeiboTech	0.409	0.27933	0.02312	0.01362	0.00000	0.00000	0.00000
Test Window W_5							
Beauty	0.904	0.00504	0.00172	0.00117	0.00000	0.00000	0.00000
Sports	0.883	0.02145	0.00651	0.00424	0.00060	0.00003	0.00003
Toys	0.925	0.01038	0.00385	0.00276	0.00000	0.00000	0.00000
Sephora	0.863	0.09608	0.19850	0.16848	0.00000	0.00000	0.00000
WikiLife	0.463	0.26678	0.20218	0.14697	0.00000	0.00000	0.00000
WeiboDaily	0.228	0.71147	0.29337	0.22213	0.00159	0.00018	0.00008
WeiboTech	0.277	0.41813	0.05636	0.03755	0.00000	0.00000	0.00000
Average							
Beauty	0.948	0.00981	0.00057	0.00039	0.00000	0.00000	0.00000
Sports	0.947	0.01072	0.00326	0.00212	0.00020	0.00001	0.00001
Toys	0.910	0.01247	0.00314	0.00172	0.00000	0.00000	0.00000
Sephora	0.773	0.11926	0.17927	0.14140	0.00000	0.00000	0.00000
WikiLife	0.569	0.31073	0.24610	0.16284	0.00000	0.00000	0.00000
WeiboDaily	0.253	0.67069	0.26844	0.20342	0.00053	0.00006	0.00003
WeiboTech	0.385	0.30972	0.02730	0.01738	0.00000	0.00000	0.00000

Table S3: Detailed @5 performance of SASRec under temporal cold-start evaluation.

Dataset	Cold ratio	Seen			Unseen		
		H@5	R@5	N@5	H@5	R@5	N@5
Test Window W_3							
Beauty	0.984	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Sports	1.000	–	–	–	0.00000	0.00000	0.00000
Toys	0.903	0.01493	0.04598	0.03188	0.00000	0.00000	0.00000
Sephora	0.634	0.04736	0.03374	0.01363	0.00000	0.00000	0.00000
WikiLife	0.650	0.00000	0.11000	0.08042	0.00000	0.00000	0.00000
WeiboDaily	0.285	0.20255	0.36725	0.30346	0.00000	0.00000	0.00000
WeiboTech	0.469	0.15854	0.04761	0.03131	0.00000	0.00000	0.00000
Test Window W_4							
Beauty	0.957	0.00000	0.00813	0.00513	0.00000	0.00000	0.00000
Sports	0.960	0.02410	0.01042	0.01042	0.00157	0.00044	0.00019
Toys	0.901	0.02252	0.00836	0.00403	0.00000	0.00000	0.00000
Sephora	0.823	0.04808	0.03557	0.01594	0.00126	0.00051	0.00026
WikiLife	0.593	0.01145	0.00730	0.00571	0.00000	0.00000	0.00000
WeiboDaily	0.247	0.23993	0.30098	0.24461	0.00000	0.00000	0.00000
WeiboTech	0.409	0.15642	0.03122	0.01987	0.00000	0.00000	0.00000
Test Window W_5							
Beauty	0.904	0.00000	0.00086	0.00041	0.00046	0.00006	0.00003
Sports	0.883	0.01125	0.00546	0.00338	0.00239	0.00006	0.00003
Toys	0.925	0.00445	0.00342	0.00211	0.00232	0.00003	0.00002
Sephora	0.863	0.01079	0.02880	0.02331	0.02556	0.00616	0.00385
WikiLife	0.463	0.00452	0.00627	0.00443	0.00000	0.00000	0.00000
WeiboDaily	0.228	0.24229	0.33069	0.27326	0.00000	0.00000	0.00000
WeiboTech	0.277	0.14912	0.03530	0.02260	0.00000	0.00000	0.00000
Average							
Beauty	0.948	0.00000	0.00300	0.00185	0.00015	0.00002	0.00001
Sports	0.947	0.01767	0.00794	0.00690	0.00132	0.00016	0.00007
Toys	0.910	0.01397	0.01925	0.01268	0.00077	0.00001	0.00001
Sephora	0.773	0.03541	0.03270	0.01763	0.00894	0.00222	0.00137
WikiLife	0.569	0.00532	0.04119	0.03019	0.00000	0.00000	0.00000
WeiboDaily	0.253	0.22826	0.33298	0.27378	0.00000	0.00000	0.00000
WeiboTech	0.385	0.15469	0.03804	0.02459	0.00000	0.00000	0.00000

Table S4: Detailed @10 performance of TIGER under temporal cold-start evaluation.

Dataset	Cold ratio	Seen			Unseen		
		H@10	R@10	N@10	H@10	R@10	N@10
Test Window W_3							
Beauty	0.984	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Sports	1.000	–	–	–	0.00000	0.00000	0.00000
Toys	0.903	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Sephora	0.634	0.28281	0.27648	0.16501	0.00000	0.00000	0.00000
WikiLife	0.650	0.38298	0.41000	0.22805	0.00000	0.00000	0.00000
WeiboDaily	0.285	0.65860	0.28977	0.19304	0.00000	0.00000	0.00000
WeiboTech	0.469	0.25610	0.00932	0.00315	0.00000	0.00000	0.00000
Test Window W_4							
Beauty	0.957	0.02439	0.00000	0.00000	0.00000	0.00000	0.00000
Sports	0.960	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Toys	0.901	0.03153	0.01393	0.00531	0.00000	0.00000	0.00000
Sephora	0.823	0.18029	0.21621	0.13985	0.00000	0.00000	0.00000
WikiLife	0.593	0.35878	0.21715	0.14444	0.00000	0.00000	0.00000
WeiboDaily	0.247	0.74306	0.35124	0.23682	0.00000	0.00041	0.00013
WeiboTech	0.409	0.36313	0.03776	0.01831	0.00000	0.00000	0.00000
Test Window W_5							
Beauty	0.904	0.01008	0.00459	0.00209	0.00023	0.00000	0.00000
Sports	0.883	0.02743	0.01155	0.00580	0.00060	0.00006	0.00004
Toys	0.925	0.01631	0.00685	0.00373	0.00000	0.00000	0.00000
Sephora	0.863	0.17361	0.24636	0.18382	0.00000	0.00000	0.00000
WikiLife	0.463	0.34122	0.26803	0.16834	0.00000	0.00000	0.00000
WeiboDaily	0.228	0.75832	0.36982	0.24692	0.00319	0.00044	0.00017
WeiboTech	0.277	0.50585	0.08658	0.04726	0.00000	0.00000	0.00000
Average							
Beauty	0.948	0.01149	0.00153	0.00070	0.00008	0.00000	0.00000
Sports	0.947	0.01371	0.00578	0.00290	0.00020	0.00002	0.00001
Toys	0.910	0.01595	0.00692	0.00301	0.00000	0.00000	0.00000
Sephora	0.773	0.21224	0.24635	0.16289	0.00000	0.00000	0.00000
WikiLife	0.569	0.36099	0.29840	0.18028	0.00000	0.00000	0.00000
WeiboDaily	0.253	0.71999	0.33694	0.22559	0.00106	0.00028	0.00010
WeiboTech	0.385	0.37502	0.04455	0.02291	0.00000	0.00000	0.00000

Table S5: Detailed @10 performance of SASRec under temporal cold-start evaluation.

Dataset	Cold ratio	Seen			Unseen		
		H@10	R@10	N@10	H@10	R@10	N@10
Test Window W_3							
Beauty	0.984	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Sports	1.000	–	–	–	0.00000	0.00000	0.00000
Toys	0.903	0.05970	0.06897	0.03904	0.00000	0.00000	0.00000
Sephora	0.634	0.04736	0.06654	0.02531	0.06304	0.03568	0.01144
WikiLife	0.650	0.04255	0.14000	0.09110	0.00000	0.00000	0.00000
WeiboDaily	0.285	0.22166	0.42399	0.32175	0.00000	0.00000	0.00000
WeiboTech	0.469	0.28049	0.07224	0.03926	0.00000	0.00000	0.00000
Test Window W_4							
Beauty	0.957	0.00000	0.02439	0.01014	0.00000	0.00000	0.00000
Sports	0.960	0.02410	0.07292	0.03017	0.00315	0.00044	0.00019
Toys	0.901	0.04505	0.03343	0.01213	0.00000	0.00000	0.00000
Sephora	0.823	0.06971	0.08182	0.03205	0.01912	0.00800	0.00259
WikiLife	0.593	0.01527	0.00912	0.00636	0.00000	0.00000	0.00000
WeiboDaily	0.247	0.25067	0.36107	0.26405	0.00207	0.00000	0.00000
WeiboTech	0.409	0.23464	0.05053	0.02614	0.00000	0.00000	0.00000
Test Window W_5							
Beauty	0.904	0.00050	0.00201	0.00077	0.00138	0.00018	0.00007
Sports	0.883	0.02286	0.00987	0.00483	0.00582	0.00011	0.00004
Toys	0.925	0.00741	0.00685	0.00323	0.00336	0.00010	0.00005
Sephora	0.863	0.07403	0.07217	0.03721	0.04006	0.01476	0.00638
WikiLife	0.463	0.00557	0.00996	0.00563	0.00039	0.00043	0.00014
WeiboDaily	0.228	0.26572	0.38945	0.29224	0.00159	0.00009	0.00003
WeiboTech	0.277	0.22515	0.05845	0.03002	0.00000	0.00000	0.00000
Average							
Beauty	0.948	0.00017	0.00880	0.00364	0.00046	0.00006	0.00002
Sports	0.947	0.02348	0.04140	0.01750	0.00299	0.00018	0.00008
Toys	0.910	0.03739	0.03641	0.01813	0.00112	0.00003	0.00002
Sephora	0.773	0.06370	0.07351	0.03152	0.04074	0.01948	0.00681
WikiLife	0.569	0.02113	0.05303	0.03437	0.00013	0.00014	0.00005
WeiboDaily	0.253	0.24602	0.39150	0.29268	0.00122	0.00003	0.00001
WeiboTech	0.385	0.24676	0.06041	0.03181	0.00000	0.00000	0.00000

Table S6: Detailed @20 performance of TIGER under temporal cold-start evaluation.

Dataset	Cold ratio	Seen			Unseen		
		H@20	R@20	N@20	H@20	R@20	N@20
Test Window W_3							
Beauty	0.984	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Sports	1.000	–	–	–	0.00000	0.00000	0.00000
Toys	0.903	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Sephora	0.634	0.43031	0.37676	0.19033	0.00000	0.00000	0.00000
WikiLife	0.650	0.38298	0.41000	0.22805	0.00000	0.00000	0.00000
WeiboDaily	0.285	0.70318	0.35125	0.20866	0.00253	0.00000	0.00000
WeiboTech	0.469	0.25610	0.01378	0.00428	0.00000	0.00000	0.00000
Test Window W_4							
Beauty	0.957	0.02439	0.00000	0.00000	0.00000	0.00000	0.00000
Sports	0.960	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Toys	0.901	0.04054	0.02786	0.00884	0.00000	0.00000	0.00000
Sephora	0.823	0.30288	0.31581	0.16471	0.00000	0.00000	0.00000
WikiLife	0.593	0.38168	0.28285	0.16094	0.00310	0.00125	0.00028
WeiboDaily	0.247	0.77708	0.41788	0.25367	0.00207	0.00101	0.00028
WeiboTech	0.409	0.40782	0.05434	0.02248	0.00000	0.00000	0.00000
Test Window W_5							
Beauty	0.904	0.01562	0.00775	0.00288	0.00069	0.00000	0.00000
Sports	0.883	0.02813	0.01702	0.00720	0.00060	0.00011	0.00005
Toys	0.925	0.02965	0.01326	0.00534	0.00026	0.00000	0.00000
Sephora	0.863	0.27835	0.31991	0.20237	0.00000	0.00000	0.00000
WikiLife	0.463	0.40870	0.32319	0.18234	0.00000	0.00000	0.00000
WeiboDaily	0.228	0.79655	0.43705	0.26397	0.00478	0.00176	0.00051
WeiboTech	0.277	0.56725	0.13624	0.05972	0.00000	0.00000	0.00000
Average							
Beauty	0.948	0.01334	0.00258	0.00096	0.00023	0.00000	0.00000
Sports	0.947	0.01406	0.00851	0.00360	0.00020	0.00004	0.00002
Toys	0.910	0.02340	0.01371	0.00473	0.00009	0.00000	0.00000
Sephora	0.773	0.33718	0.33749	0.18581	0.00000	0.00000	0.00000
WikiLife	0.569	0.39112	0.33868	0.19044	0.00103	0.00042	0.00009
WeiboDaily	0.253	0.75894	0.40206	0.24210	0.00313	0.00092	0.00026
WeiboTech	0.385	0.41039	0.06812	0.02883	0.00000	0.00000	0.00000

Table S7: Detailed @20 performance of SASRec under temporal cold-start evaluation.

Dataset	Cold ratio	Seen			Unseen		
		H@20	R@20	N@20	H@20	R@20	N@20
Test Window W_3							
Beauty	0.984	0.25000	0.00000	0.00000	0.00000	0.00000	0.00000
Sports	1.000	–	–	–	0.00000	0.00000	0.00000
Toys	0.903	0.08955	0.10345	0.04738	0.00000	0.00000	0.00000
Sephora	0.634	0.06089	0.08529	0.02965	0.06304	0.03568	0.01144
WikiLife	0.650	0.06383	0.15000	0.09355	0.03448	0.03226	0.00759
WeiboDaily	0.285	0.25096	0.48114	0.33620	0.00000	0.00000	0.00000
WeiboTech	0.469	0.43902	0.10504	0.04755	0.00000	0.00000	0.00000
Test Window W_4							
Beauty	0.957	0.00000	0.03252	0.01206	0.00253	0.00074	0.00018
Sports	0.960	0.07229	0.12500	0.04304	0.00472	0.00044	0.00019
Toys	0.901	0.05405	0.05014	0.01638	0.00000	0.00000	0.00000
Sephora	0.823	0.08053	0.09960	0.03622	0.03571	0.01448	0.00423
WikiLife	0.593	0.04580	0.03467	0.01253	0.00000	0.00000	0.00000
WeiboDaily	0.247	0.26321	0.42386	0.27989	0.00414	0.00000	0.00000
WeiboTech	0.409	0.28492	0.07552	0.03244	0.00000	0.00000	0.00000
Test Window W_5							
Beauty	0.904	0.00101	0.00258	0.00092	0.00988	0.00129	0.00035
Sports	0.883	0.02707	0.01387	0.00584	0.00776	0.00022	0.00007
Toys	0.925	0.01334	0.01412	0.00503	0.00439	0.00010	0.00005
Sephora	0.863	0.07875	0.07743	0.03846	0.07120	0.02582	0.00917
WikiLife	0.463	0.01148	0.01568	0.00704	0.00039	0.00043	0.00014
WeiboDaily	0.228	0.30086	0.45330	0.30837	0.00159	0.00009	0.00003
WeiboTech	0.277	0.27778	0.09432	0.03901	0.00000	0.00000	0.00000
Average							
Beauty	0.948	0.08367	0.01170	0.00433	0.00414	0.00068	0.00018
Sports	0.947	0.04968	0.06943	0.02444	0.00416	0.00022	0.00009
Toys	0.910	0.05232	0.05590	0.02293	0.00146	0.00003	0.00002
Sephora	0.773	0.07339	0.08744	0.03478	0.05665	0.02532	0.00828
WikiLife	0.569	0.04037	0.06678	0.03771	0.01163	0.01090	0.00258
WeiboDaily	0.253	0.27167	0.45276	0.30815	0.00191	0.00003	0.00001
WeiboTech	0.385	0.33391	0.09163	0.03967	0.00000	0.00000	0.00000

Table S8: Token-level coldness taxonomy for unseen target items. The table separates cold items according to token support and prefix support.

Dataset	Coldness category	Ratio	TIGER					
			Recall@5	Recall@10	Recall@20	NDCG@5	NDCG@10	NDCG@20
Beauty	all-token-seen	0.521	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	any-token-unseen	0.479	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1	0.954	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1, c_2	0.173	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1, c_2, c_3	0.001	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Sports	all-token-seen	0.405	0.00001	0.00003	0.00006	0.00001	0.00002	0.00003
	any-token-unseen	0.595	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1	0.723	0.00001	0.00002	0.00004	0.00001	0.00001	0.00002
	prefix-seen-c_1, c_2	0.208	0.00002	0.00004	0.00008	0.00002	0.00003	0.00004
	prefix-seen-c_1, c_2, c_3	0.002	0.00336	0.00671	0.01007	0.00336	0.00437	0.00520
Toys	all-token-seen	0.713	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	any-token-unseen	0.287	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1	1.000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1, c_2	0.452	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1, c_2, c_3	0.004	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Sephora	all-token-seen	0.475	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	any-token-unseen	0.525	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1	1.000	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1, c_2	0.650	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1, c_2, c_3	0.002	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
WikiLife	all-token-seen	0.824	0.00000	0.00000	0.00045	0.00000	0.00000	0.00010
	any-token-unseen	0.176	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1	0.977	0.00000	0.00000	0.00042	0.00000	0.00000	0.00010
	prefix-seen-c_1, c_2	0.304	0.00000	0.00000	0.00175	0.00000	0.00000	0.00040
	prefix-seen-c_1, c_2, c_3	0.001	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
WeiboDaily	all-token-seen	1.000	0.00006	0.00028	0.00092	0.00003	0.00010	0.00026
	any-token-unseen	0.000	–	–	–	–	–	–
	prefix-seen-c_1	1.000	0.00006	0.00028	0.00092	0.00003	0.00010	0.00026
	prefix-seen-c_1, c_2	0.516	0.00009	0.00038	0.00138	0.00004	0.00014	0.00039
	prefix-seen-c_1, c_2, c_3	0.010	0.00080	0.00320	0.01386	0.00034	0.00114	0.00379
WeiboTech	all-token-seen	0.999	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	any-token-unseen	0.001	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1	0.999	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1, c_2	0.591	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
	prefix-seen-c_1, c_2, c_3	0.008	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000

Table S9: Oracle-prefix probing for semantic ID generation. We report both next-item semantic ID generation accuracy and next-token semantic ID generation accuracy. "Free" denotes standard autoregressive decoding, while "Oracle-prefix" decoding forces the corresponding ground-truth prefix before generating the remaining tokens. Δ denotes the absolute improvement over free decoding. Darker red indicates larger improvement.

Dataset	Group	Ratio	Next-item SID generation					Next-token SID generation						
			Free	Oracle c_1		Oracle c_1, c_2		c_1 Acc.	c_2 Acc.			c_3 Acc.		
				Acc.	Δ	Acc.	Δ	Free	Free	Oracle c_1	Δ	Free	Oracle c_1, c_2	Δ
Beauty	Seen	0.052	0.00010	0.00823	0.00813	0.02383	0.02373	0.07487	0.02465	0.03639	0.01174	0.00163	0.02421	0.02259
	Unseen	0.948	0.00000	0.00000	0.00000	0.00134	0.00134	0.02181	0.00778	0.00870	0.00092	0.00126	0.00140	0.00014
Sports	Seen	0.053	0.00063	0.00630	0.00567	0.02815	0.02752	0.18009	0.04449	0.07936	0.03487	0.00284	0.02910	0.02626
	Unseen	0.947	0.00001	0.00006	0.00005	0.00448	0.00447	0.11983	0.00494	0.01091	0.00597	0.00138	0.00530	0.00392
Toys	Seen	0.090	0.00057	0.00564	0.00506	0.09664	0.09607	0.25815	0.01563	0.04140	0.02577	0.00514	0.09814	0.09300
	Unseen	0.910	0.00000	0.00000	0.00000	0.00272	0.00272	0.19895	0.00587	0.01855	0.01268	0.00391	0.00334	-0.00057
Sephora	Seen	0.227	0.10155	0.15851	0.05696	0.69089	**0.58933**	0.38365	0.12973	0.21057	0.08084	0.11470	0.69117	**0.57648**
	Unseen	0.773	0.00000	0.00000	0.00000	0.00063	0.00063	0.29701	0.01904	0.05075	0.03171	0.00679	0.00072	-0.00606
WikiLife	Seen	0.431	0.09446	0.16480	0.07034	0.50373	**0.40927**	0.53002	0.12913	0.18835	0.05922	0.09624	0.50379	**0.40755**
	Unseen	0.569	0.00000	0.00000	0.00000	0.00308	0.00308	0.27493	0.00845	0.01816	0.00971	0.00384	0.00322	-0.00062
WeiboDaily	Seen	0.747	0.12319	0.47262	**0.34943**	0.75122	**0.62804**	0.18646	0.13814	0.49332	**0.35518**	0.12957	0.75163	**0.62206**
	Unseen	0.253	0.00003	0.00049	0.00046	0.01493	0.01490	0.06264	0.01797	0.02774	0.00977	0.01480	0.01972	0.00492
WeiboTech	Seen	0.615	0.00632	0.07472	0.06841	0.39882	**0.39251**	0.08608	0.01498	0.10570	0.09071	0.01139	0.39933	**0.38794**
	Unseen	0.385	0.00000	0.00004	0.00004	0.01173	0.01173	0.06236	0.00881	0.01749	0.00868	0.00647	0.01666	0.01020

Table S10: Distributional alignment between TIGER prefix generation and SASRec bucket-aggregated scoring. Each row compares the TIGER prefix distribution $P_T(\cdot)$ with the SASRec bucket distribution $P_S(\cdot)$ at the corresponding semantic-ID prefix level. Lower JS divergence and higher Pearson correlation indicate stronger alignment.

Metric	SID prefix	Beauty	Sports	Toys	Sephora	WikiLife	WeiboDaily	WeiboTech
JS divergence ↓	c_1	0.12821	0.29669	0.25995	0.09539	0.19576	0.20870	0.24888
	c_1, c_2	0.49832	0.53745	0.56818	0.33208	0.36511	0.39852	0.56114
	c_1, c_2, c_3	0.64622	0.63961	0.66619	0.47234	0.40976	0.46563	0.60842
Pearson ↑	c_1	0.58776	0.46533	0.44644	0.63721	0.58386	0.53238	0.36585
	c_1, c_2	0.39636	0.29548	0.18975	0.42604	0.39078	0.36548	0.11598
	c_1, c_2, c_3	0.19580	0.00009	0.00480	0.29056	0.28029	0.16506	0.10909

Table S11: Extended session-wise Hit@20 on three kinds of methods on seven datasets. Bold denotes the best result for each dataset and target group.

Method	Category	Target	Beauty	Sports	Toys	Sephora	WikiLife	WeiboDaily	WeiboTech
TIGER	SID Generators	Seen	0.01334	0.01406	0.02340	**0.33718**	0.39112	0.75894	0.41039
		Unseen	0.00023	0.00020	0.00009	0.00000	0.00103	0.00313	0.00000
TIGER-SID		Seen	0.00655	0.01582	0.02790	0.32466	0.38787	0.77940	**0.49199**
		Unseen	0.00054	0.00109	0.00000	0.00000	0.00013	0.00747	0.01232
TIGER-Edge		Seen	0.01925	0.03886	**0.08387**	0.31388	**0.43409**	**0.81308**	0.47153
		Unseen	0.00042	0.00075	0.00000	0.00198	0.00026	0.01132	0.01683
RPG	Alternative Generative Designs	Seen	0.01741	0.03564	0.07459	0.31118	0.39963	0.74209	0.42838
		Unseen	0.00061	0.00085	0.00009	0.01796	0.00129	0.00983	0.01530
SpecGR		Seen	0.01453	0.02861	0.05390	0.24219	0.30920	0.57841	0.33539
		Unseen	0.00130	0.00222	0.00009	0.04695	0.00339	**0.02317**	**0.03882**
Latte		Seen	0.01692	0.03502	0.06820	0.30159	0.38935	0.72627	0.41955
		Unseen	0.00046	0.00062	0.00009	0.01303	0.00079	0.00704	0.01073
TIGER-Scorer	Candidate Scorers	Seen	0.02163	0.04120	0.00920	0.24545	0.28630	0.32633	0.16490
		Unseen	**0.00831**	**0.01392**	**0.01187**	0.00003	0.00505	0.00750	0.02929
SASRec		Seen	**0.08367**	**0.04968**	0.05232	0.07339	0.04037	0.27167	0.33391
		Unseen	0.00414	0.00416	0.00146	**0.05665**	**0.01163**	0.00191	0.00000

Table S12: Extended tokenizer and prediction factorization session-wise Hit@20 on seven datasets. Bold denotes the best result for each dataset and target group.

Tokenizer	Prediction	Target	Beauty	Sports	Toys	Sephora	WikiLife	WeiboDaily	WeiboTech
RQ-VAE	AR	Seen	0.01334	0.01406	0.02340	**0.33718**	**0.39112**	0.75894	0.41039
		Unseen	0.00023	0.00020	0.00009	0.00000	**0.00103**	0.00313	0.00000
PQ	AR	Seen	**0.01348**	0.01324	0.02317	0.32458	0.37902	0.73462	0.40285
		Unseen	0.00008	0.00005	0.00000	0.00000	0.00026	0.00053	0.00000
OPQ	AR	Seen	0.00655	**0.01582**	**0.02790**	0.32466	0.38787	**0.77940**	**0.49199**
		Unseen	0.00054	0.00109	0.00000	0.00000	0.00013	0.00747	0.01232
RQ-VAE	Parallel	Seen	0.00549	0.00519	0.00903	0.13412	0.15380	0.30152	0.16239
		Unseen	0.00000	0.00000	0.00000	0.00000	0.00013	0.00027	0.00000
PQ	Parallel	Seen	0.01269	0.01372	0.02149	0.32162	0.37245	0.72775	0.38832
		Unseen	0.00008	0.00005	0.00000	0.00000	0.00026	0.00053	0.00000
OPQ	Parallel	Seen	0.01320	0.01362	0.02316	0.32783	0.37591	0.73158	0.39855
		Unseen	**0.00065**	**0.00129**	**0.00017**	**0.00932**	0.00026	**0.00865**	**0.01427**